

----- Slip 2 -----

Q.1) Write a C program to find file properties such as inode number, number of hard link, File permissions, File size, File access and modification time and so on of a given file using stat() system call.

```
#include <stdio.h>
#include <sys/stat.h>
#include <pwd.h>
#include <grp.h>
#include <time.h>

void print_permissions(mode_t mode) {
    printf((S_ISDIR(mode)) ? "d" : "-");
    printf((mode & S_IRUSR) ? "r" : "-");
    printf((mode & S_IWUSR) ? "w" : "-");
    printf((mode & S_IXUSR) ? "x" : "-");
    printf((mode & S_IRGRP) ? "r" : "-");
    printf((mode & S_IWGRP) ? "w" : "-");
    printf((mode & S_IXGRP) ? "x" : "-");
    printf((mode & S_IROTH) ? "r" : "-");
    printf((mode & S_IWOTH) ? "w" : "-");
    printf((mode & S_IXOTH) ? "x" : "-");
}

int main(int argc, char *argv[]) {

    if (argc != 2) {
        printf("Usage: %s <filename>\n", argv[0]);
        return 1;
    }

    struct stat fileStat;

    if (stat(argv[1], &fileStat) < 0) {
        perror("stat");
    }
}
```

```

    return 1;
}

printf("\n--- File Properties for: %s ---\n\n", argv[1]);

printf("Inode Number    : %lu\n", fileStat.st_ino);
printf("Number of Links  : %lu\n", fileStat.st_nlink);

printf("File Permissions  : ");
print_permissions(fileStat.st_mode);
printf("\n");

printf("File Size      : %ld bytes\n", fileStat.st_size);

printf("UID          : %u (%s)\n",
      fileStat.st_uid, getpwuid(fileStat.st_uid)->pw_name);

printf("GID          : %u (%s)\n",
      fileStat.st_gid, getgrgid(fileStat.st_gid)->gr_name);

printf("Last Access Time : %s", ctime(&fileStat.st_atime));
printf("Last Modify Time : %s", ctime(&fileStat.st_mtime));

// Detect file type
printf("File Type      : ");
if (S_ISREG(fileStat.st_mode)) printf("Regular File\n");
else if (S_ISDIR(fileStat.st_mode)) printf("Directory\n");
else if (S_ISCHR(fileStat.st_mode)) printf("Character Device\n");
else if (S_ISBLK(fileStat.st_mode)) printf("Block Device\n");
else if (S_ISFIFO(fileStat.st_mode)) printf("FIFO/Named Pipe\n");
else if (S_ISLNK(fileStat.st_mode)) printf("Symbolic Link\n");
else if (S_ISSOCK(fileStat.st_mode)) printf("Socket\n");
else printf("Unknown\n");

```

```
    return 0;
}
```

Q.2) Write a C program that catches the ctrl-c (SIGINT) signal for the first time and display the appropriate message and exits on pressing ctrl-c again.

```
#include <stdio.h>
#include <stdlib.h>
#include <signal.h>

int count = 0;

// Signal handler function
void sigint_handler(int sig) {
    count++;

    if (count == 1) {
        printf("\nCtrl-C detected. Press Ctrl-C again to exit.\n");
    } else {
        printf("\nSecond Ctrl-C detected. Exiting now.\n");
        exit(0);
    }
}

int main() {

    // Register the signal handler for SIGINT
    signal(SIGINT, sigint_handler);

    printf("Program running... Press Ctrl-C.\n");

    // Infinite loop to keep the program running
    while (1) {
        // Do nothing, just wait for signals
    }
}
```

```
    return 0;  
}
```